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with limited intercellular spaces. They contain numerous chloroplasts.

Vascular Bundles

Vascular bundles differ in size. Most of the vascular bundles are smaller in size. Large bundles occur at regular intervals.Two patches of sclerenchyma are present above and below the large vascular bundles. These sclerenchyma patches give mechanical support to the leaf. The small vascular bundles do not have such sclerenchymatous patches. Vascular bundles are **conjoint, collateral and closed**. Each vascular bundle is surrounded by a parenchymatous bundle sheath. The cells of the bundle sheath generally contain starch grains. The xylem of the vascular bundle is located towards the upper epidermis and the phloem towards the lower epidermis.In C_4 grasses, the bundle sheath cells are living and involve in C_4 photosynthesis. This sheath is called **Kranz sheath**.

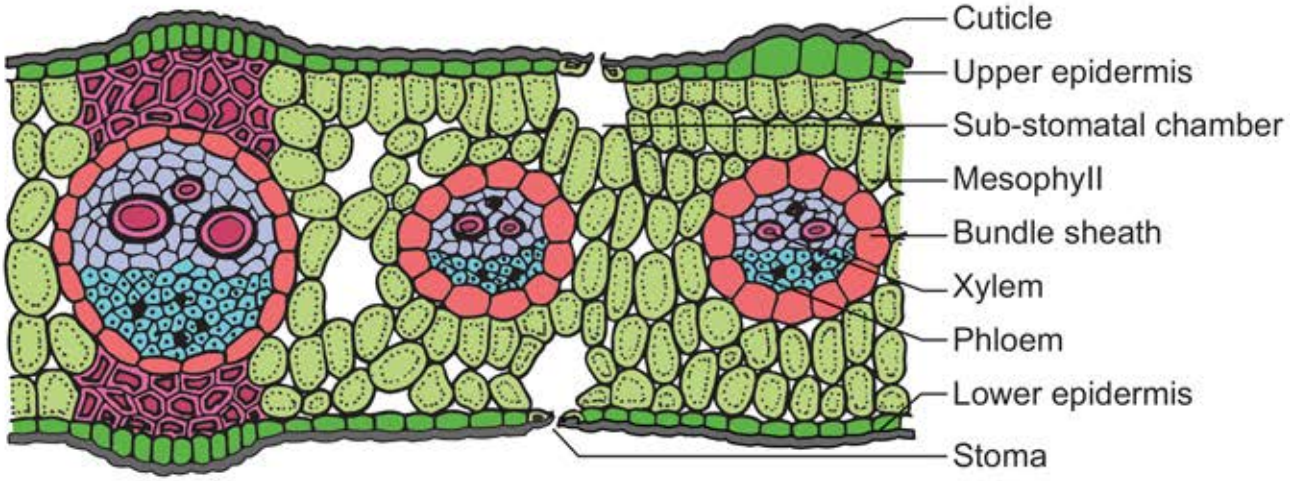


Figure 9.25: T.S. of monocot leaf (Grass)

Water Stomata (or) Hydathodes

A **hydathode** is a type of epidermal pore, commonly found in higher plants.

Structurally, hydathodes are modified stomata, usually located at leaf tips or margins, especially at the teeth.

Hydathodes occur in the leaves of submerged aquatic plants such as **Ranunculus fluitans** as well as in many herbaceous land plants.

Hydathodes are made of a group of living cells with numerous intercellular spaces filled with water, but few or no chloroplasts. These cells open out into one or more sub-epidermal chambers. These, in turn, communicate with the exterior through an open pore. The water stoma structurally resembles an ordinary stoma, but is usually larger and has lost the power of movement.They are connected to the plant vascular system by a tracheid or vessel element.

Hydathodes discharge liquid water with various dissolved substances from the interior of the leaf to its surface. This process is called **guttation**. Example many grasses.